A Literacy-Free, Report-Free Eye Exam on a Consumer EEG Headband and a Consumer VR Headset: Toward Objective Vision Screening by Community Health Workers

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***Abstract*—At least a billion people live with a vision impairment that could be prevented or corrected, uncorrected refractive error first among the causes, and the bottleneck is not the lens but the screen: an eye test today needs a literate, cooperating subject and a trained examiner, and both are scarce where the need is greatest. We describe an eye exam that asks nothing of the subject but to look at a dot with one eye covered, on two consumer devices that together cost under 600 US dollars: a Muse headband with one electrode added over the visual cortex, and a Meta Quest 3S whose browser renders the stimulus, with the subject's phone already in the loop as the sensor hub. Four tests run as ON/OFF blocks of flicker, a grating on its own mean grey with bars shrinking (acuity) or contrast falling (contrast), one sector of a 4×4 grid at a time (field), and the patch in red, green and blue (colour); each step is scored by how much of the flicker frequency the EEG carries against the test's OFF blocks, with an exact permutation test, and thresholds follow the sweep-VEP rule used for infants, who cannot report either. A hardware-free phantom with set cut-offs, a dead sector and colour gains passes all twelve checks of a full-stack gate. The first sitting on a person fitted no threshold: both ear electrodes had come loose under the headset strap and a signal check built for electrodes at the amplifier rails had passed them. That failure produced what a device in a health worker's hands must have, a contact check that tells the worker what to fix before each eye. The same recording carries the blink rate and the occipital rhythm that our companion sittings read as eye strain, which puts a just-in-time "rest your eyes" prompt within reach of the same headband. Nothing here is a diagnosis; it is the apparatus, its validation, and an honest first sitting.**

***Index Terms*—mobile health; visual screening; SSVEP; sweep VEP; wearable sensors; consumer electronics; extended reality; community health workers; just-in-time adaptive intervention.**

I. Introduction: the screen, not the lens, is the bottleneck

The World Health Organization counts at least 2.2 billion people with a vision impairment, at least a billion of them with one that could have been prevented or has yet to be addressed, and names uncorrected refractive error the leading cause [1]. The remedy is a pair of glasses; the gap is finding the people who need them. A letter chart needs a subject who can read it, or name a tumbling E, in a language the examiner shares, and an examiner who can judge the answers; the objective alternatives, autorefractors and clinical visual evoked potentials, cost tens of thousands of dollars and stay in the clinic. The pattern that has carried other screens to the community health worker is a consumer device, a worker, and software that checks itself: a phone that screens for pulmonary disease [2], and before it the case for low-cost wearable sensors as the instruments of frontline care [3]. We follow that pattern for the eye. The steady-state visual evoked potential (SSVEP), the rhythm the visual cortex produces in time with a flickering stimulus, has a clinical pedigree as an objective acuity test: the sweep VEP follows the response to a grating as its bars get finer and takes the acuity where the response is extrapolated to vanish, and it was built for infants, who cannot report [4]; the same logic gives an electrophysiological contrast threshold [5]. An objective test also answers a fairness question before it is asked: it does not depend on literacy, on language, or on how the examiner hears an answer [6]. It brings biases of its own, hair and skin and the fit of a headband, and we meet one of them in Section V.

II. The rig: consumer hardware, the phone as hub

Mann et al. showed that the SSVEP from a few scalp electrodes can turn the eye into a camera, a square flickering at one position after another and the response saying how bright each position was [7]. We rebuilt that rig on consumer hardware (Fig. 2a): a Muse headband, validated for event-related potentials against laboratory EEG [8], with one cup electrode added at Oz over the visual cortex; the subject's phone, which receives the headband over Bluetooth and streams raw EEG to a laptop over Wi-Fi, the architecture of wireless autonomic sensing that iCalm set out [9] and that consumer wrist sensors later made ordinary [10]; and a Meta Quest 3S whose browser renders the stimulus frame-exactly and talks the subject through the test (Fig. 1a). On that rig we have read words, eight colours and a sealed optotype from one subject's EEG. The headband and the headset together cost under 600 US dollars, and nothing in the exam needs the laptop to be more than a phone with a screen.

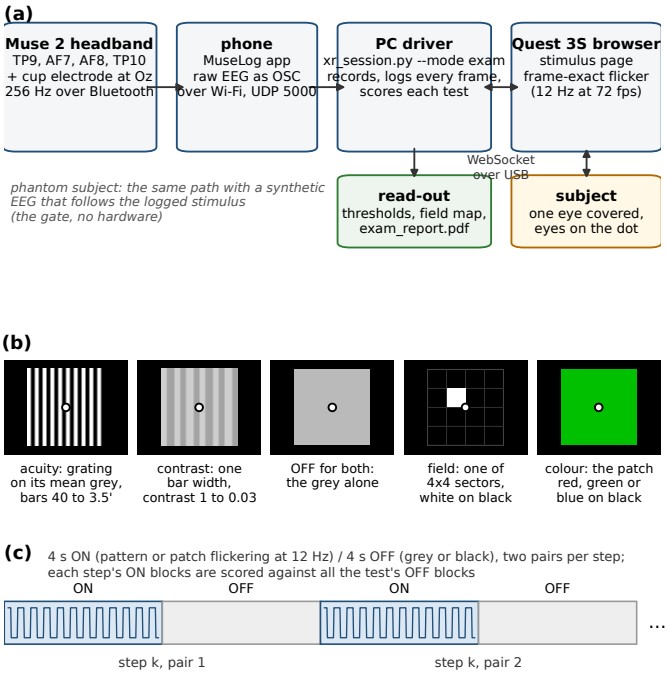


Fig. 1. The exam. (a) The path from scalp to read-out: the headband streams raw EEG through the phone to the PC, which drives the stimulus page in the headset's browser over USB, logs every frame it draws, and scores each test; the phantom subject replaces the first two boxes with a synthetic EEG that follows the logged stimulus. (b) What the headset shows in the four tests, and the OFF screen of the grating tests. (c) The block timeline of one step.

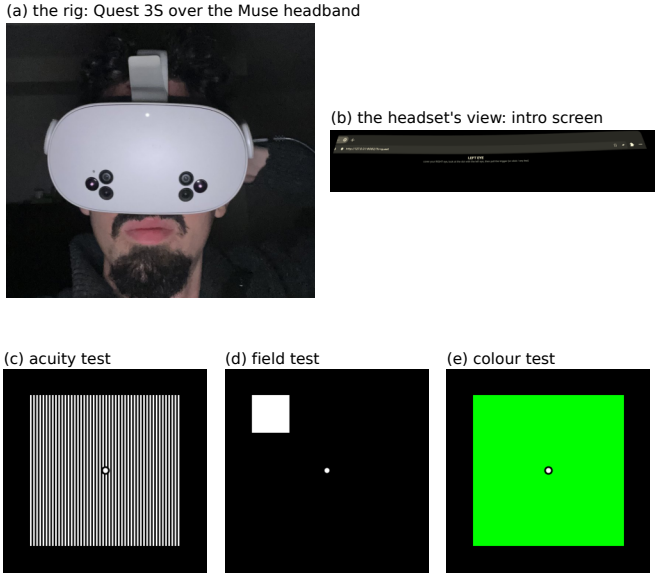


Fig. 2. The rig and the page. (a) The Quest 3S worn over the Muse headband (the cup electrode at Oz is under the strap). (b) The headset's own view of the intro screen of one eye, captured from the device. (c)–(e) The stimulus page during an ON frame of the acuity, field and colour tests, captured from the browser: the grating on its own grey, one white sector of the 4×4 grid, the green plane, each with the fixation dot.

III. The Exam

The subject looks at a central dot with one eye covered while the page runs each test as ON/OFF blocks (4 s each, two pairs per step) at one flicker frequency, frame-exact on the headset's 72 Hz display (12 Hz, three frames on and three off); Fig. 1 shows the stimuli and the timeline and Fig. 2 the page as rendered. For acuity a vertical black/white grating appears and disappears on a grey of the same mean luminance, so only the pattern comes and goes, its bar width stepping from 40 to 3.5 arcmin; for contrast the grating keeps one bar width while its Michelson contrast steps from 1 to 0.03; for the field one sector of a 4×4 grid flickers white against black while the eyes stay on the dot; for colour the whole patch flickers red, green or blue. Bar widths are set in arcmin from the display's pixels per degree, rounded to whole device pixels, and the width delivered is reported with every block. Every frame and block edge is logged, and each step is scored on the channel with the largest ON/OFF difference over the test: line power at the flicker frequency and its double against the flanking spectrum as a log signal-to-noise ratio per block, the step's ON blocks against all the test's OFF blocks by an exact permutation test over the block labels, a noise-corrected amplitude, and the response phase against the logged flicker edges (relative between sectors and eyes, not a latency). A channel whose rms stays above 60 μV through a test is set aside as out of contact. Thresholds follow the sweep-VEP rule [4]: from the step just below the finest step that still responds, through it and up the rising limb, a line is fitted and extrapolated to zero amplitude; a threshold is refused when only one step responds, and one below the range shown is said so rather than turned into a Snellen value. The field is reported as amplitude and phase per sector with the silent sectors named and, with two eyes, their sector-by-sector ratio and phase difference; colour as the amplitude of each plane. The whole exam is about seven minutes per eye, and its output is one picture.

IV. Validation without a patient

A device meant for a worker without an eye clinic behind them has to be testable without a patient in front of it. The rig is therefore gated against a phantom subject that reads the stimulus log as it is written and streams synthetic EEG whose SSVEP follows what is on screen at the frequency actually delivered. For the exam the phantom's eye got a logistic visibility in log bar width and log contrast with a cut-off per eye, sectors that give no response, and a gain per colour plane. The gate dials in a right eye with cut-offs of 8 arcmin and contrast 0.08 and one dead sector, a left eye twice as coarse and whole, and gains of 0.7, 1 and 0.4 for red, green and blue, runs the whole stack (driver, headless browser, phantom; 3 s blocks, 13.7 minutes) and asks twelve things: every stage delivered, each fitted threshold inside the band where the phantom's visibility runs from 0.03 to 0.6 (a line through the rising limb crosses zero below the half-visibility point), the left eye at least $1.3\times$ coarser on both thresholds, exactly the dead sector named, and green $\geq$ red $>$ blue in both eyes. All twelve pass (Fig. 3): right eye 3.9 arcmin and 0.031 with sector six silent, left eye 7.8 arcmin and 0.067 with every sector responding, and the response phase within 42 to 47° in every responding sector of both eyes and random in the

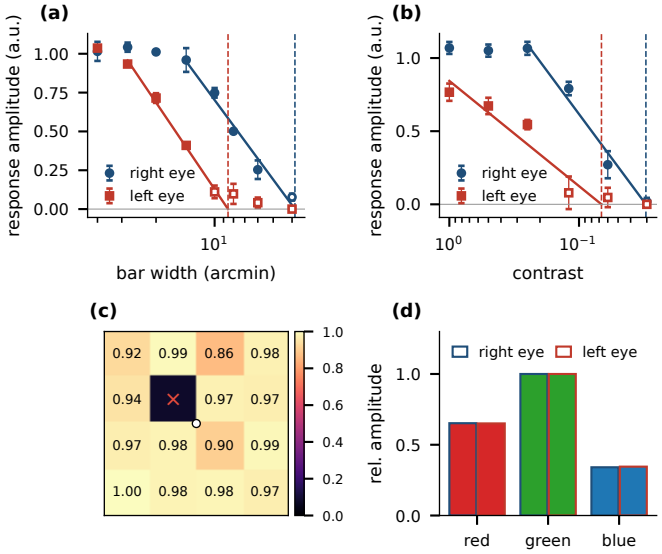


Fig. 3. The phantom gate's read-out. (a) Acuity and (b) contrast: noise-corrected amplitude at the flicker frequency and its double per bar width and per contrast for the right (circles) and left (squares) phantom eye; open markers are not significant ($p \geq 0.05$); lines are the fitted limbs, dashed verticals their zero crossings. (c) The right eye's field: amplitude per sector relative to the best, the cross the sector the phantom cannot see, the dot the fixation point. (d) Colour planes of both eyes relative to the best plane.

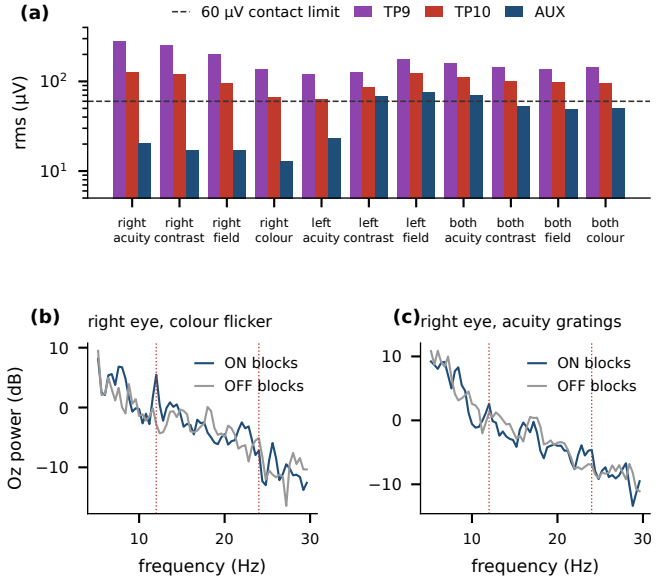


Fig. 4. The first sitting's raw EEG. (a) Rms of the two ear electrodes and of Oz through every test of the two runs against the 60 μV contact limit: the ears never held, Oz held for the right eye only. (b) Oz spectrum in the ON blocks against the OFF blocks of the right eye's colour test, with the 12 Hz line and its double marked: the one clear response of the sitting. (c) The same for the right eye's acuity test.

silent one. A first run of the gate had lost one contrast step to noise and fitted the left eye's contrast from the saturating top of its curve, four times too low; that is why the fit anchors on the step below the finest responding one, as the sweep method does by spanning the fall to the noise floor. The gate runs unattended in a quarter of an hour, which is what lets the exam be changed without a volunteer for every change.

V. The First Sitting, and What a Field Device Must Do About It

The first sitting on a person (one of the authors) ran the right eye with the left covered, then the left, and a second run with both eyes open, at 12 Hz on the Quest, in one evening. No threshold could be fitted. The only clear response was the full-patch colour flicker of the right eye, a 12 Hz line on Oz at a $\log_{10}$ signal-to-noise ratio of $+0.75$ in ON blocks against -0.22 in OFF blocks (Fig. 4b); the gratings gave a weak response at the coarsest bars (Fig. 4c), the sectors and contrast steps none. The raw EEG says why (Fig. 4a): both ear electrodes sat at 100 to 280 μV rms for the whole sitting, loose under the headset strap, while the signal check that opens every session had reported them good because it was written to catch an electrode floating at the amplifier rails and accepts anything below 350 μV; the Oz electrode was clean for the right eye (13 to 18 μV) and rose to 50 to 77 μV afterwards; and the patch was 0.6 of a browser window that was not full screen. This is the failure a consumer wearable brings to the field: not a wrong number but a quiet one, an electrode that has moved a few millimetres [3], [8], and a worker without an oscilloscope cannot see it. The sitting changed the apparatus more than the phantom had: before each eye the exam now measures every channel's rms and, when any is above 60 μV, shows an electrode check on the headset that names the electrode and what to do (press the ear pads in, wet the top electrode) and waits for a trigger pull until every channel holds; the scorer sets a channel out of contact aside; the read-out names it, and refuses to extrapolate a single responding step into a number; the page goes full screen on the exam's own trigger pull with a patch the full height of the window. Hair, head shape and the fit of a headband are also where an objective test can be unfair in its own way, and the contact check is the first of the audits that a deployment would need [6].

VI. What the Same Recording Carries

An eye exam on a headband records more than the visual response, and the rest of the recording is the kind of signal affective computing has drawn on since its start [11]. In our companion sittings on the same rig (in preparation) the forehead electrodes gave the blink rate, which stops while a patch flickers and bursts the second it goes dark, and which doubled between two sittings of one evening while a first daytime sitting stayed at the subject's resting rate; the occipital rhythm drifted with the electrodes rather than with the flicker, which is why the exam reads nothing into it. Blink rate is an old marker of fatigue and of eye strain [12], [13], and it comes for free from the same two minutes that give a threshold. That is the hook for a just-in-time adaptive intervention [14]: the device that has just measured an eye can also notice that the eyes are tired and say so, a "rest your eyes" prompt that costs nothing to deliver and is the smallest intervention a screen can carry. We say estimate, not detection, and eye strain, not

emotion; what the recording supports is an activation reading against the subject's own earlier minutes.

VII. Toward the field

What is established is the apparatus and its validation: four tests per eye that run without a response from the subject on a headband and a headset sold to consumers, a scorer with an exact test per step, a threshold rule taken from the sweep VEP, and a phantom gate that catches the failures a human sitting is too slow and too dear to find. What is not established is any measurement on a person; the first sitting is reported as the reason the contact check exists, not as a result. Two things are open before a second sitting can mean anything: the display's pixels per degree, which sets the arcmin scale and has not been measured on the headset (the comparison between the eyes does not need it, a Snellen value does), and the flicker rate for the pattern tests, since pattern onset at 12 Hz shows the grating for three frames at a time and a slower frame-exact rate such as 6 Hz may drive the response that the luminance flicker of the colour test already drives. After that the road is the one every screen has walked: a blur study with trial lenses against a letter chart to tie the extrapolated bar width to acuity, a normative cohort, and a worker-facing flow in which the phone starts the stream, the headset talks the subject through, the device checks its own contact, and the report is one picture a worker can act on and a clinic can read. The headband's sensitivity to an electrode that has moved is the practical limit of the consumer wearable; a screen built on it must check its own contact at every step, and this one now does.

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